👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 0 Shop Start-Up & Safety (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
His readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as his worksheets for that lesson
- Quick check: score, then scale to his test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Shop Safety TEST**: The Canvas quiz has 0 questions in the export; our tests are team-written.
- **Room-specific rules**: Extinguisher, eyewash and exit locations depend on Mr. Davis's shop; add them on the shop walk.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

🛡️ Shop Start-Up & Safety: Guided Notes

Small Engines Overhaul · Unit 0 · Level 3 (closer to the class work) · goes with the syllabus, shop contract, class norms and Shop Safety TEST

Name: _____

Use the class slides and your reading. Key words:

PPE | *safety glasses* | *coveralls* | *flammable* | *vapor* | *carbon monoxide* | *ventilation* | *fire extinguisher* | *spark plug wire*

A. PPE

B. Fire and air

2. Gasoline vapors can ignite from a _____.

3. Oily rags go in the closed metal safety _____.

4. Why must engines run outdoors or with ventilation?

_____

_____

C. Getting into the shop

5. List the syllabus requirements for shop work.

_____

_____

🛡️ Shop Start-Up & Safety: Reading

Small Engines Overhaul · Unit 0 · Level 3 (closer to the class work) · goes with the syllabus, shop contract, class norms and Shop Safety TEST

Name: _____

Working Safely in the Small Engines Shop

Mr. Davis's class is a hands-on shop class. Each student tears down and rebuilds a small gas engine. Before you can work in the shop, you must sign the syllabus and shop contract with a parent or guardian, finish the class assignments, and pass the shop safety test with at least 80 percent.

Personal protective equipment, or PPE, protects your body. Safety glasses are worn the entire time you are in the shop because parts, springs, dirt and fluids can fly without warning. Coveralls keep oil and grease off your clothes. Closed-toe shoes protect your feet from dropped parts. Long hair is tied back and loose jewelry is removed so nothing catches in moving parts.

Fuel is the biggest fire risk. Gasoline is flammable, and its vapors can ignite from a single spark. Fuel is drained with adult supervision in a ventilated area, and oily rags go in the closed metal safety can. Everyone should know where the fire extinguisher and exits are.

Running engines make exhaust that contains carbon monoxide, a deadly gas you cannot see or smell, so engines are only run outdoors or with good ventilation. Before working on an engine, disconnect the spark plug wire so it cannot start by accident. At the end of every class, return each tool to its spot, wipe up spills and sweep your area.

Check your reading

1. What three things must you do before working in the shop?

2. Why are safety glasses worn the whole time?

3. Why disconnect the spark plug wire before working?

🧯 Shop Safety Walk & Tool Board: Lab Sheet

Small Engines Overhaul · Unit 0 · Level 3 · goes with Class Norms / FHS shop contract

Name: _____

Question: Where are the safety equipment, exits and hazards in our shop?

Materials

shop map sheet, safety glasses, clipboard

⚠️ Safety

- Safety glasses and coveralls on
- Look only; do not touch tools or engines
- Stay with the group

Prediction

Where do you predict the fire extinguisher will be, and why?

_____

_____

Procedure

1. Walk the shop with an adult.
2. Mark extinguisher, eyewash, first aid and exits.
3. Mark fuel cabinet and oily-rag can.
4. List 5 tools from the tool board with their uses.
5. Compare maps with a partner.

Data

Safety item	Location	Why it is there
Fire extinguisher		
Eyewash		
Fuel cabinet		
Oily-rag can		

Analysis (Claim, Evidence, Reasoning)

1. Claim: which hazard in our shop is the most serious? Give evidence from your map and explain how the shop controls it.

🛡️ Shop Start-Up & Safety: Quick Check

Small Engines Overhaul · Unit 0 · Level 3 (closer to the class work) · goes with the syllabus, shop contract, class norms and Shop Safety TEST

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. When do you wear safety glasses?

A. Only when grinding | B. Only outside | C. The whole time in the shop

2. Where do oily rags go?

A. The closed metal safety can | B. The trash can | C. A cardboard box

3. Carbon monoxide is…

A. A kind of fuel | B. A cleaning spray | C. A deadly, odorless gas in exhaust

4. Before working on an engine you…

A. Disconnect the spark plug wire | B. Fill the tank | C. Start it

5. Describe two fuel safety rules.

6. Why does the class require 80% on the safety test before shop work?

🛡️ Shop Start-Up & Safety: Guided Notes · ANSWER KEY

Small Engines Overhaul · Unit 0 · Level 3 (closer to the class work) · goes with the syllabus, shop contract, class norms and Shop Safety TEST

Name: _____

Use the class slides and your reading. Key words:

PPE | *safety glasses* | *coveralls* | *flammable* | *vapor* | *carbon monoxide* | *ventilation* | *fire extinguisher* | *spark plug wire*

A. PPE

B. Fire and air

2. Gasoline vapors can ignite from a **spark** .

3. Oily rags go in the closed metal safety **can** .

4. Why must engines run outdoors or with ventilation?
Exhaust has carbon monoxide, a deadly gas you cannot see or smell.

C. Getting into the shop

5. List the syllabus requirements for shop work.
Signed syllabus and contract, all class assignments done, safety test passed with 80%.

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

🧯 Shop Safety Walk & Tool Board: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 0 · Level 3 · goes with Class Norms / FHS shop contract

Name: _____

Question: Where are the safety equipment, exits and hazards in our shop?

Materials

shop map sheet, safety glasses, clipboard

⚠️ Safety

- Safety glasses and coveralls on
- Look only; do not touch tools or engines
- Stay with the group

Prediction

Where do you predict the fire extinguisher will be, and why?
Near an exit and near the fuel area, easy to reach.

Procedure

1. Walk the shop with an adult.
2. Mark extinguisher, eyewash, first aid and exits.
3. Mark fuel cabinet and oily-rag can.
4. List 5 tools from the tool board with their uses.
5. Compare maps with a partner.

Data

Safety item	Location	Why it is there
Fire extinguisher	**near exit**	**fight small fires fast**
Eyewash	**near sink**	**flush eyes**
Fuel cabinet	**away from heat**	**store fuel safely**
Oily-rag can	**by benches**	**prevent rag fires**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: which hazard in our shop is the most serious? Give evidence from your map and explain how the shop controls it.
Example: fuel; the cabinet and safety can keep it away from sparks.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

🛡️ Shop Start-Up & Safety: Quick Check · ANSWER KEY

Small Engines Overhaul · Unit 0 · Level 3 (closer to the class work) · goes with the syllabus, shop contract, class norms and Shop Safety TEST

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. When do you wear safety glasses?

A. Only when grinding	B. Only outside	**C. The whole time in the shop**

2. Where do oily rags go?

A. The closed metal safety can	B. The trash can	C. A cardboard box

3. Carbon monoxide is…

A. A kind of fuel	B. A cleaning spray	**C. A deadly, odorless gas in exhaust**

4. Before working on an engine you…

A. Disconnect the spark plug wire	B. Fill the tank	C. Start it

5. Describe two fuel safety rules.

No sparks/flames near fuel; drain fuel with adult supervision in ventilation; oily rags in safety can.

6. Why does the class require 80% on the safety test before shop work?

To prove you know the rules so you and others stay safe around tools, fuel and engines.

🔑 *Team-written from the syllabus and shop rules; the real Shop TEST is not in the export.*